

Resume
Yohei Ohata

Email: yohei.ohata.ee@hitachi.com

Education

The University of Tokyo, Japan April 2022—March 2024

M.S., Department of Human and Engineered Environmental Sciences

***Summa cum laude* Dean's Award for Outstanding Achievement**

Cumulative GPA: 4.0/4.0

Master's Thesis: Proposal of a Thought Process Analysis Method for Identifying Different Comprehension Methods for Learners.

Tokyo University of Science, Japan April 2018—March 2022

B.Eng., Applied Electronics

Cumulative GPA: 3.5/4.0

Undergraduate Thesis: A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions.

Publications

Articles in Peer-Reviewed Journals

1. **Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (submitted). Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models. *Human Factors*.

Peer-Reviewed Conference Proceedings

1. **Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2024). Visualizing and Revealing the Difference of Learner's Thought Process and Question Solving Method. *12th International Conference on Information and Education Technology (ICIET)*, Yamaguchi, Japan, pp. 472-476. doi:10.1109/ICIET60671.2024.10542759
2. **Ohata, Y.**, Takao, K., Natori, T., & Aikawa, N. (2021). A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions. *IEICE Tech. Rep.*, 121(196), pp. 12-16.

Conference Presentations

1. **Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2023). Analysis of How Learners Learn Using Graphic Questions. *The 42nd Annual Conference of Japanese Society for Educational Technology*, Tokyo, Japan, pp.357-358.
2. **Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2022). Identifying Perceptual Factors that Influence Impression of a Teacher in a Classroom. *The 41st Annual Conference of Japanese Society for Educational Technology*, Kanagawa, Japan, pp.197-198.

Patents

1. **Ohata, Y.** & Jin, Y. (2025). Management Issues Estimation Device. Japan Patent Japan Patent and Trademark Office.

Research Experience

Research Assistant, Dr. [Amano Lab](#) at The University of Tokyo May 2025—Present

- Conducting data-driven analysis to identify shared brain states across diverse cognitive tasks.
- Conducted preprocessing and confounds regression of BOLD data using fMRIPrep and Nilearn.
- GLM-based first- and second-level analyses.
- Developed temporal alignment techniques across participants using “BrainSync” (MATLAB).
- Conducted tensor decomposition.
- Network composition ratio analysis using Yeo's 7 networks.
- Developed containerized workflows for HPC-based computation.

Research Assistant, Dr. [Hachisuka Lab](#) at The University of Tokyo April 2022—March 2024

- Designed experiments and built experimental environments.
- Conducted studies involving both adults and elementary school students.
- Measured physiological signals, with a focus on eye-tracking data.
- Analyzed physiological data using Python.
- Developed machine learning models (hidden Markov models) based on eye-tracking data.
- Conducted interviews and administered questionnaire surveys.
- Performed statistical analyses to interpret experimental results.
- Developed a web-based UI and operation log acquisition system using HTML, CSS, and JavaScript.
- Authored and revised research articles.

Research Assistant, Dr. [Aikawa Lab](#) at Tokyo University of Science April 2022—March 2024

- Conducted signal processing simulations and performance evaluations.
- Developed and implemented mathematical models.
- Programmed and analyzed data using MATLAB.

Work Experience

Hitachi, Ltd., Japan April 2024—Present

Researcher, Digital Innovation R&D

- Developed concepts and use cases for new services to support business creation.
- Defined functional requirements and evaluated appropriate methodologies.
- Authored and submitted patent applications.
- Explored the application of Large Language Models (LLMs) in service development.
- Established conditions and evaluation criteria for technical verification of LLMs.

Teaching Experience

The University of Tokyo, Japan April 2023—June 2023

Teaching Assistant: Measurement Basics II

- Assisted in building signal processing systems using LabVIEW.

Gakuho Seminar, Japan

April 2019—March 2022

Teacher

- Taught Mathematics, Physics, Chemistry, and English to students in Years 10–18.
- Created and managed individualized study plans to support academic progress.
- Successfully guided over 30 students in passing their target school entrance examinations.

HONORS & AWARDS

IEEE CAS JJC Best Student Award

2021

Language Skills

Programing: Python/Matlab/Docker/JavaScript/HTML/CSS/Bash/SSH&HPC

High level in English

Native in Japanese; Nationality: Japanese

References

Satori Hachisuka, Ph.D.

Associate Professor, Applied Computer Science Course, Interfaculty Initiative in Information Studies

The University of Tokyo

E-mail: hachisuka@edu.k.u-tokyo